Applied Network Security Analysis

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Executive Summary

Colonial Pipeline Company operates the largest refined oil pipeline in the United States. The system spans over 5,500 miles and transports gasoline, diesel, jet fuel, and other refined petroleum products from the Gulf Coast to New Jersey, supplying about 45 percent of the fuel consumed on the East Coast. As a critical infrastructure provider, any disruption to its operations can have wide societal and economic effects. In May 2021, the company fell victim to a ransomware attack that forced a complete shutdown of operations for six days, the longest outage in its history. The attackers stole approximately 100 GB of sensitive data, encrypted key IT systems, and disabled the billing platform, effectively halting fuel distribution as a result of compromised IT security protocols. Following the attack several mitigation strategies were implemented such as access control, network security , response and recovery and regular security audits.

1.0 Introduction

The Colonial Pipeline ransomware attack was a significant cybersecurity incident that occurred in **May 2021** and affected the largest fuel pipeline operator in the United States, which supplies nearly half of the East Coast's gasoline, diesel, and jet fuel. The attack was carried out by a criminal ransomware group known as **DarkSide**, which gained access to the company's information technology systems and encrypted critical data, demanding a ransom for its release.

As a precautionary measure, Colonial Pipeline shut down its pipeline operations to prevent further spread of the attack, leading to temporary fuel shortages and disruptions across several states. Although the operational control systems were not directly compromised, the incident demonstrated how cyberattacks on corporate IT environments can have serious real-world consequences for national critical infrastructure. It also drew global attention to the need for stronger cybersecurity controls, incident response planning, and regulatory oversight in the energy sector.

2.0 Aim and objectives

The aim of this lab is to critically review, analyse, and elaborate the ransomware attack incident while proposing an industry standard mitigation for similar attacks as future reference for other businesses in the energy sector.

The Objectives of this lab is to

- analyse the attack vector
- analyse the compromised security architecture
- analyse the incident response preparedness
- analyse the corresponding effects of compromised security systems on business operations
- proposed a solid mitigation plan

2.1 Tools

- Open Source research

3.0 Methodology

3.1 Compromised Cybersecurity Architecture

3.1.1 The Initial Attack Vector

Initial access was gained through an unused Legacy VPN account that relied on a single factor authentication and had never been deactivated. Its credentials had been exposed in a previous data breach and were later found in a public password dump. The account did not use multi-factor authentication, meaning a single username and password provided direct access to the network. During the COVID-19 pandemic, remote access capabilities had been expanded without equivalent increases in monitoring and security controls. As a result, suspicious activities such as privilege escalation and lateral movement went undetected for days.

3.1.2 Key vulnerability

The success of the attack was largely due to several underlying weaknesses. Authentication controls were inadequate, with no multi-factor authentication and outdated VPN infrastructure. Network segmentation was insufficient, allowing attackers to move laterally between IT environments leading to the shutdown of operational technology by Colonial pipelines management for threat containment. The organization lacked modern detection tools such as EDR and SIEM platforms, and monitoring capabilities were limited. These gaps allowed the attackers to operate with little resistance.

3.1.3 The Threat Type (Ransomware Attack)

Once inside, the attackers deployed a customized variant of the DarkSide ransomware, designed for both encryption and data theft in a double-extortion model. They used legitimate administrative tools such as PowerShell, along with malicious frameworks like Cobalt Strike to establish command-and-control channels and persistent access. Tools such as Mimikatz were used to extract credentials and escalate privileges. They performed reconnaissance, mapped the network, staged data for exfiltration, and employed anti-forensic techniques such as log clearing and “living off the land” methods to evade detection. The attackers stole approximately 100 GB of sensitive data, encrypted key IT systems, and disabled the billing platform

3.2 Target Infrastructure

3.2.1 IT Business Network

The attackers initially compromised Colonial Pipeline’s corporate IT network. This environment hosted employee systems, email, authentication services, and business applications. The ransomware encrypted critical servers and data in this network, disrupting internal operations and prompting the company to shut down interconnected systems as a precaution.

3.2.2 Billing System

Colonial’s billing and invoicing systems, which rely on the IT network, were rendered unavailable. Although the attackers did not directly target the physical pipeline controls, the loss

of billing capability meant the company could not accurately track fuel deliveries and payments, making it unsafe to continue operations.

3.2.3 Operational Technology Impact

The operational impact was severe. Following the shutdown, fuel shortages spread across 17 states, prices surged, and panic buying led to empty stations and long queues. Airlines experienced jet fuel shortages and rerouted flights. Supply chains for essential goods were disrupted. Colonial paid a ransom of approximately $4.4 million in Bitcoin, part of which was later recovered by the FBI through blockchain tracing. Although operations resumed by May 13, full recovery took weeks

3.3 Protocol Failure and Relevance

3.3.1 Authentication Protocol Failure (VPN and MFA)

The initial compromise occurred through a legacy VPN account that relied only on single-factor authentication (username and password). The credentials were likely reused from a previous breach and were no longer actively monitored.

Without a robust authentication protocol such as Multi-Factor Authentication (MFA), possession of valid credentials was sufficient for access.

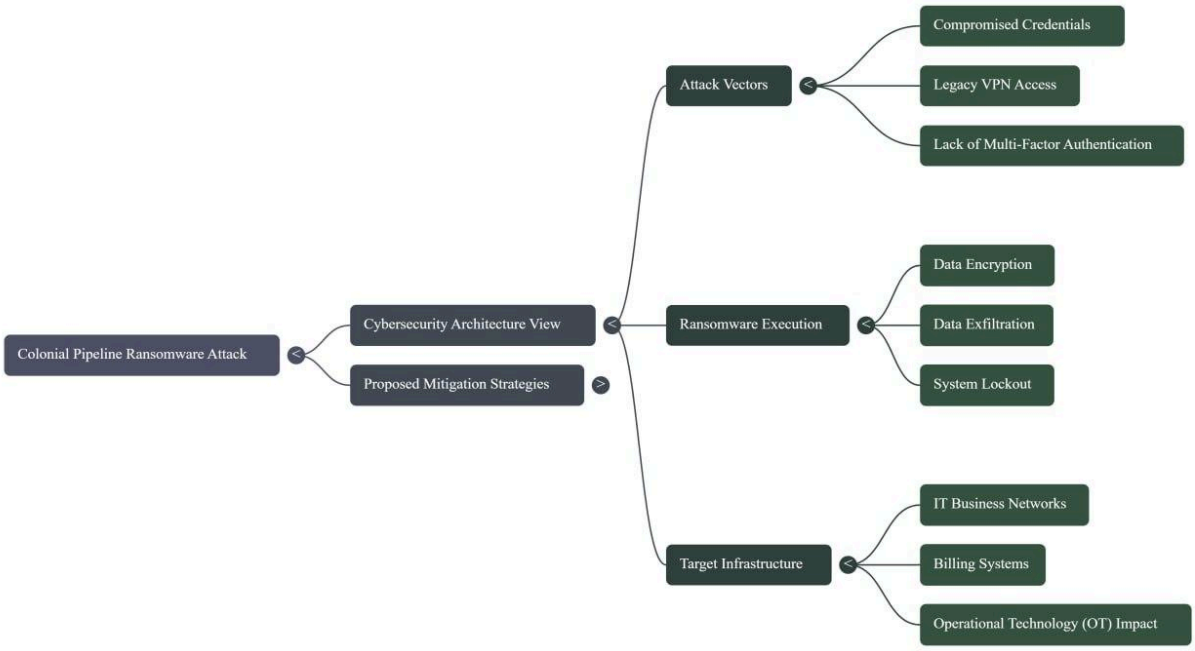
3.3.2 Network Segmentation (IT and OT Separation)

Network segmentation is the practice of dividing a network into isolated security zones using firewalls, VLANs, and access control policies. In critical infrastructure environments, this typically means strict separation between:Although Colonial's OT environment was not directly infected, the lack of sufficiently strong logical and operational segmentation created uncertainty about whether the ransomware could propagate from IT to OT. The company could not confidently guarantee that the control network was isolated.

As a result, they shut down pipeline operations as a defensive measure.

3.3.3 Encryption Protocols (SSL/TLS and IPSec)

Protocols such as SSL/TLS and IPSec are designed to protect data in transit by providing confidentiality, integrity, and authentication of communication channels. In the Colonial Pipeline case, these protocols were not the root cause of the breach.



3.4 Part 3: Mitigation Strategy for the Colonial Pipeline Attack

3.4.1 Access Control

- MFA
- Zero trust authentication
- Privilege Access Management

Security Control:

Multi-Factor Authentication (MFA) with Zero Trust Architecture and Privileged Access Management (PAM)

How it Addresses the Vulnerability:

The 2021 attack began with the compromise of a VPN account protected only by a password.
Enforcing MFA would have required an additional verification factor, preventing unauthorized access even if credentials were stolen.
Zero Trust authentication ensures that no user or device is automatically trusted, even after initial login, and continuous verification is required.
Privileged Access Management restricts and monitors high-level accounts, ensuring attackers cannot escalate privileges easily after entry.

Justification (Best Practice):

Industry standards such as NIST SP 800-63 and CISA Zero Trust guidelines recommend MFA and least-privilege enforcement for all remote access and administrative accounts. These controls significantly reduce credential-based attacks, which remain the leading cause of ransomware intrusions.

3.4.2 Network Security

- Network Segmentation
- IT/OT Air Gapping
- IDS

Security Control:

Network Segmentation with IT/OT Air-Gapping and Intrusion Detection Systems (IDS)

How it Addresses the Vulnerability:

Weak segmentation between the IT business network and operational systems forced Colonial Pipeline to shut down operations due to uncertainty about ransomware spread.
Strong segmentation using firewalls, VLANs, and access control lists, combined with physical or logical air gaps between IT and OT, would prevent lateral movement.
IDS would detect abnormal traffic and ransomware behavior early, allowing isolation of infected segments before business-wide shutdown.

Justification (Best Practice):
IEC 62443 and NIST SP 800-82 recommend strict zoning and conduit models for industrial control systems, separating corporate and control networks. Real-time monitoring with IDS is a core requirement for detecting intrusions in critical infrastructure environments.

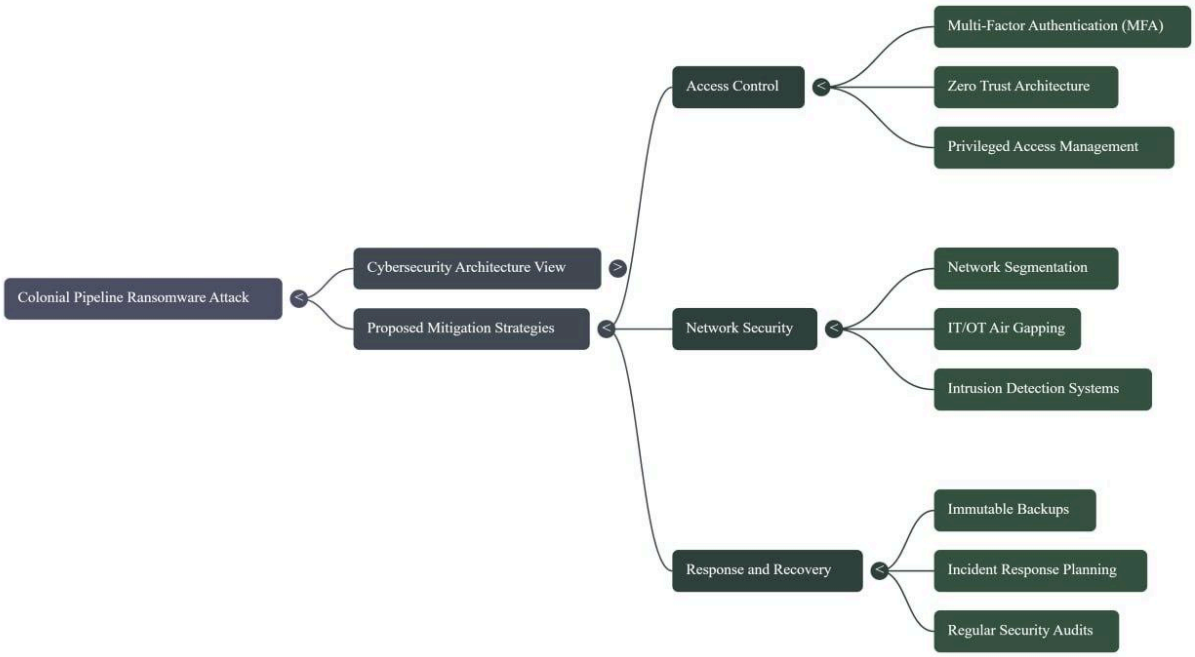
3.4.3 Response and Recovery

- Immutable Backups
- Incident Response Planning
- Regular Security Audit

Security Control:
Immutable Backups, Incident Response Planning, and Regular Security Audits

How it Addresses the Vulnerability:
Ransomware encrypted business systems and disrupted operations. Immutable backups stored offline or in write-once-read-many (WORM) storage would allow rapid restoration without paying ransom.
A formal incident response plan ensures rapid containment, communication, and decision-making during breaches.
Regular security audits identify outdated VPN accounts, missing MFA enforcement, and misconfigurations before attackers exploit them.

Justification (Best Practice):
NIST SP 800-61 and ISO 27035 emphasize tested incident response procedures and resilient backup strategies as core components of cyber resilience. Immutable backups are now a standard ransomware defense recommended by CISA, ENISA, and major cloud security frameworks.



4.0 Conclusion

The Colonial Pipeline ransomware attack marked a turning point in national cybersecurity awareness. It demonstrated that cyber incidents can cause real-world disruption on a massive scale and that critical infrastructure is a prime target for financially motivated attackers. The event exposed weaknesses, but it also accelerated long-overdue reforms in policy, technology, and cooperation. It was not just an isolated breach, but a watershed moment that reshaped how nations and organizations think about protecting the systems that underpin modern society.

The group responsible was DarkSide, a ransomware-as-a-service syndicate that leased its tools to affiliates in exchange for a share of the profits. The group operated like a business, maintained a public-facing presence on the dark web, and avoided targeting systems using Russian language settings. After the Colonial incident drew intense global attention, DarkSide claimed the attack was conducted by an affiliate and later announced it was shutting down, though similar groups often reappear under new names.